

Multivariable Calculus and Differential Equations

An intuitive, proof-based course from geometry to computation

Functions of many variables - partial and total derivatives - Taylor theory - multiple integrals
- substitution - integration by parts - implicit functions - first- and second-order ODEs -
Euler's method

Designed as self-contained lecture notes with worked examples, proofs, exercises, and
complete solutions

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How to use this book

This text is written for a reader who does not want a bag of formulas. The goal is to make the formulas feel inevitable. Each major topic follows the same cycle:

1. identify the geometric or physical question;
2. define the mathematical object precisely;
3. derive the computational rule;
4. prove the rule under clean hypotheses;
5. practice on examples from easy to difficult;
6. test understanding with exercises, then compare against complete solutions.

The two halves of the book are tightly connected. Multivariable calculus studies how a quantity changes when there are several independent directions. Differential equations reverse that viewpoint: they prescribe the local change and ask you to reconstruct the function or trajectory.

A central theme will appear repeatedly:

local linear information $\longrightarrow$ global approximation or evolution.

Notation and prerequisites

A point in the plane is written $\mathbf{x} = (x, y)$. A point in $\mathbb{R}^n$ is $\mathbf{x} = (x_1, \dots, x_n)$. Vectors are bold when clarity helps. The Euclidean norm is

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \dots + x_n^2}.$$

For a scalar function $f(x, y)$, we write $f_x = \partial f / \partial x$, $f_y = \partial f / \partial y$, and

$$\nabla f = (f_x, f_y).$$

You should be comfortable with one-variable algebra, trigonometry, exponentials, logarithms, elementary derivatives, and basic definite integrals. Everything else needed is developed here.

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Part I

Functions of Several Variables

From a Line to a Landscape

1.1 What is a multivariable function?

A one-variable function $y = f(x)$ receives one input and returns one output. A multivariable function receives several inputs. The basic two-variable form is

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (x, y) \mapsto f(x, y).$$

You can imagine (x, y) as a location on a map and $f(x, y)$ as the temperature, elevation, pressure, cost, probability density, or error measured at that location.

Intuition

A graph $y = f(x)$ is a curve because one input coordinate plus one output coordinate makes a two-dimensional picture. A graph $z = f(x, y)$ is a surface because two input coordinates plus one output coordinate make a three-dimensional picture. The function is not the surface itself; the surface is one visualization of the rule.

Examples:

$$f(x, y) = x^2 + y^2, \quad T(x, y) = 20 + 3x - y, \quad P(V, T) = \frac{nRT}{V}.$$

The first is a paraboloid, the second is a plane, and the third is a physical law involving two independent variables when n, R are fixed.

1.2 Domain, range, graph, and level sets

The *domain* is the set of allowed inputs. For

$$f(x, y) = \sqrt{4 - x^2 - y^2},$$

we need $4 - x^2 - y^2 \geq 0$, so the domain is the closed disk $x^2 + y^2 \leq 4$.

The graph is

$$\{(x, y, z) : z = f(x, y)\}.$$

A *level set* fixes the output:

$$f(x, y) = c.$$

For $f(x, y) = x^2 + y^2$, the level sets are circles $x^2 + y^2 = c$. For $f(x, y) = x^2 + 2y^2$, they are ellipses.

Intuition

A topographic map is made of level curves. Walking along a level curve changes your position but not your altitude. Later, this will explain why the gradient is perpendicular to level curves.

1.3 Linear and affine functions in two variables

A linear map $L : \mathbb{R}^2 \rightarrow \mathbb{R}$ has the form

$$L(x, y) = ax + by.$$

An affine function adds a constant:

$$f(x, y) = ax + by + c.$$

Its graph is a plane

$$z = ax + by + c.$$

The key point is not just that planes are easy. Linear maps are the objects that best describe small changes. Multivariable differentiation is built around this fact.

1.4 Quadratic functions in two variables

A general quadratic polynomial is

$$Q(x, y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F.$$

The homogeneous quadratic part can be written using a symmetric matrix:

$$Ax^2 + Bxy + Cy^2 = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

That matrix will later become the Hessian in the second-order Taylor approximation.

Positive definite quadratic form

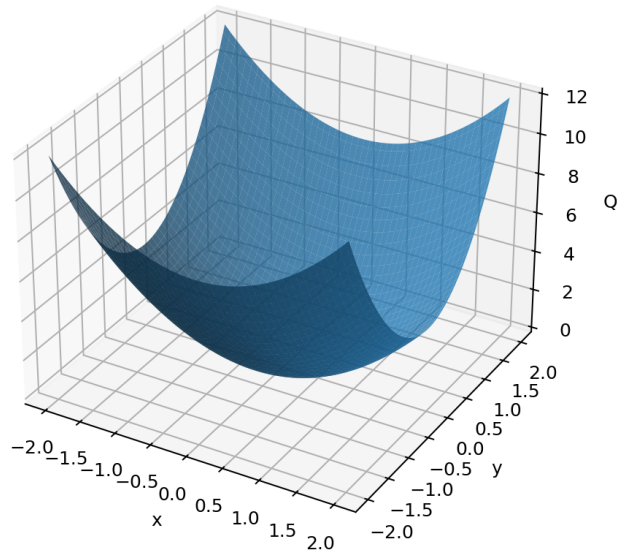


Figure 1.1: A positive definite quadratic form has a bowl shape.

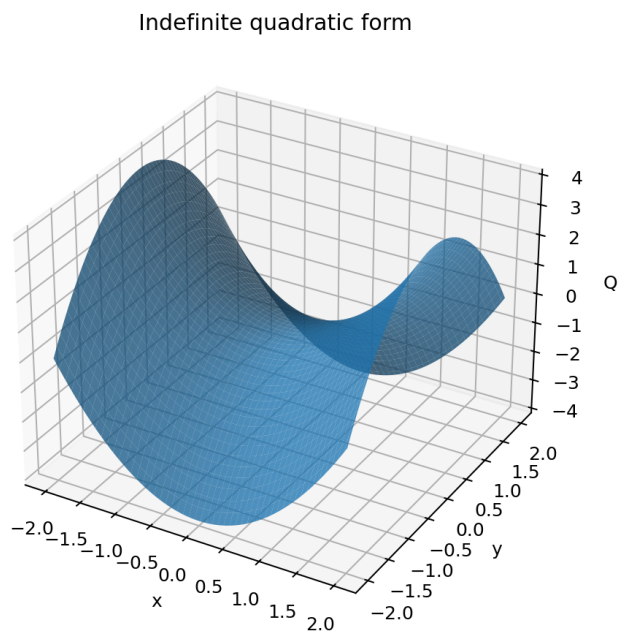


Figure 1.2: An indefinite quadratic form rises in some directions and falls in others.

Exercises

1. Find the domain of $f(x, y) = \log(9 - x^2 - y^2)$.
2. Describe the level curves of $f(x, y) = x^2 + 4y^2$.
3. Write $3x^2 + 4xy + 5y^2$ as $\mathbf{x}^T A \mathbf{x}$ with symmetric A .
4. Classify the shape of $Q(x, y) = x^2 - y^2$ by looking along the lines $y = 0$ and $x = 0$.

Limits and Continuity in Several Variables

2.1 Why multivariable limits are harder

In one variable, $x \rightarrow a$ approaches from two sides. In two variables, $(x, y) \rightarrow (a, b)$ can approach along infinitely many paths: lines, parabolas, spirals, or arbitrary curves.

Definition 2.1. We say

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \implies |f(x, y) - L| < \varepsilon.$$

This definition is path-independent by construction: every sufficiently nearby point must work.

2.2 Path tests: useful for disproving, not proving

Consider

$$f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}.$$

Along $y = 0$, $f = 1$. Along $x = 0$, $f = -1$. Therefore no limit exists at the origin.

But checking ten or a thousand paths cannot prove a limit exists; there are infinitely many paths. To prove existence, bound the function by something depending only on the distance to the target point.

Worked example

Prove

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3}{x^2 + y^2} = 0.$$

Since $x^2 \leq x^2 + y^2$,

$$\left| \frac{x^3}{x^2 + y^2} \right| = |x| \frac{x^2}{x^2 + y^2} \leq |x| \leq \sqrt{x^2 + y^2}.$$

If $\sqrt{x^2 + y^2} < \delta$, the absolute value is $< \delta$. Choose $\delta = \varepsilon$.

2.3 Continuity

A function is continuous at (a, b) when

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

Polynomials are continuous everywhere. Rational functions are continuous wherever the denominator is nonzero. Compositions of continuous functions are continuous where defined.

Theorem 2.2 (Sequential criterion). *f is continuous at $\mathbf{a}$ if and only if for every sequence $\mathbf{x}_n \rightarrow \mathbf{a}$, we have $f(\mathbf{x}_n) \rightarrow f(\mathbf{a})$.*

Proof

If f is continuous, then for any $\varepsilon > 0$ choose δ from continuity. Since $\mathbf{x}_n \rightarrow \mathbf{a}$, eventually $\|\mathbf{x}_n - \mathbf{a}\| < \delta$, hence $|f(\mathbf{x}_n) - f(\mathbf{a})| < \varepsilon$.

Conversely, suppose continuity fails. Then there exists $\varepsilon_0 > 0$ such that for every n we can find $\mathbf{x}_n$ with $\|\mathbf{x}_n - \mathbf{a}\| < 1/n$ but $|f(\mathbf{x}_n) - f(\mathbf{a})| \geq \varepsilon_0$. Then $\mathbf{x}_n \rightarrow \mathbf{a}$ while the function values do not converge to $f(\mathbf{a})$, contradiction.

Exercises

1. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2+y^2}} = 0$.
2. Show that $\frac{xy}{x^2+y^2}$ has no limit at $(0,0)$.
3. Determine whether $f(x, y) = \sqrt{1 - x^2 - y^2}$ is continuous at every point of its domain.

Part II

Multivariable Differentiation

Partial Derivatives: Looking One Direction at a Time

3.1 Definition

For $z = f(x, y)$, freeze y and vary x . The partial derivative is

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}.$$

Similarly,

$$f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h}.$$

Intuition

Imagine standing on a mountain. f_x is the slope if you walk exactly east-west while keeping the north-south coordinate fixed. f_y is the slope if you walk exactly north-south. They are slices through the surface, not yet the full story of how the surface changes in an arbitrary direction.

Worked example

For

$$f(x, y) = x^2y + e^{xy},$$

we get

$$f_x = 2xy + ye^{xy}, \quad f_y = x^2 + xe^{xy}.$$

When taking f_x , treat y as a constant; when taking f_y , treat x as a constant.

3.2 Higher partial derivatives

Differentiate again:

$$f_{xx}, \quad f_{xy}, \quad f_{yx}, \quad f_{yy}.$$

For sufficiently smooth functions, mixed partials agree.

Theorem 3.1 (Clairaut-Schwarz theorem). *If f_{xy} and f_{yx} are continuous in a neighborhood of (a, b) , then*

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Proof

For small h, k , define the rectangular increment

$$\Delta = f(a + h, b + k) - f(a + h, b) - f(a, b + k) + f(a, b).$$

Apply the one-variable mean value theorem first in x to the function $x \mapsto f(x, b + k) - f(x, b)$. For some ξ between a and $a + h$,

$$\Delta = h [f_x(\xi, b + k) - f_x(\xi, b)].$$

Apply the mean value theorem in y to $f_x(\xi, y)$. For some η between b and $b + k$,

$$\Delta = hk f_{xy}(\xi, \eta).$$

Repeating the argument in the reverse order gives points ξ', η' with

$$\Delta = hk f_{yx}(\xi', \eta').$$

Therefore $f_{xy}(\xi, \eta) = f_{yx}(\xi', \eta')$. As $h, k \rightarrow 0$, all intermediate points approach (a, b) , and continuity gives the result.

3.3 Partial derivatives do not automatically imply continuity

A common mistake is to think: “both partial derivatives exist, therefore the function is differentiable.” False.

Define

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Along the axes, $f = 0$, so $f_x(0, 0) = f_y(0, 0) = 0$. But along $y = x$, $f = 1/2$, so f is not even continuous at the origin.

Exercises

1. Compute all first and second partial derivatives of $f = x^3y^2 + \sin(xy)$.
2. Verify $f_{xy} = f_{yx}$ for $f = e^{x+2y} \cos(xy)$.
3. Construct the tangent slopes in the x and y coordinate slices for $f = x^2 + 2y^2$ at $(1, -1)$.

The Total Derivative: The Real Multi-variable Derivative

4.1 Why partial derivatives are not enough

For one variable, differentiability means

$$f(a + h) = f(a) + f'(a)h + o(|h|).$$

The derivative is the best linear approximation.

In two variables, the increment is a vector $\mathbf{h} = (h, k)$. The correct generalization is not merely two numbers; it is a linear map.

Definition 4.1. $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is differentiable at $\mathbf{a} = (a, b)$ if there exists a linear map $L : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + L(\mathbf{h}) + r(\mathbf{h}),$$

where

$$\frac{r(\mathbf{h})}{\|\mathbf{h}\|} \rightarrow 0 \quad \text{as } \mathbf{h} \rightarrow 0.$$

The map L is the total derivative $Df(\mathbf{a})$.

Every linear map $L : \mathbb{R}^2 \rightarrow \mathbb{R}$ has the form

$$L(h, k) = Ah + Bk.$$

For a differentiable function, $A = f_x(a, b)$ and $B = f_y(a, b)$. Hence

$$Df(a, b)(h, k) = f_x(a, b)h + f_y(a, b)k.$$

4.2 The tangent plane

Writing $z = f(x, y)$, the linearization at (a, b) is

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

This is the tangent plane.

Surface and tangent plane

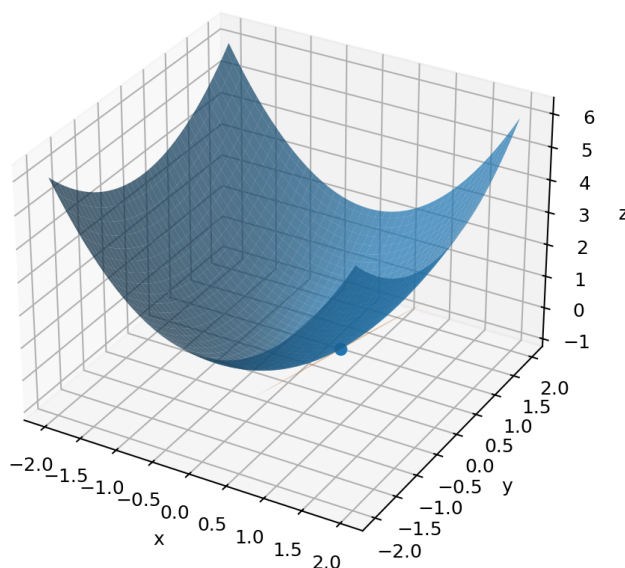


Figure 4.1: Differentiability means that after zooming in, the surface becomes indistinguishable from its tangent plane.

Intuition

Differentiability is a statement about error. The tangent plane is not merely touching the surface; its error is much smaller than the distance moved. If you move a distance r , a differentiable function's linearization error is $o(r)$.

Theorem 4.2 (Continuous partials imply differentiability). *If f_x and f_y exist in a neighborhood of (a, b) and are continuous at (a, b) , then f is differentiable at (a, b) .*

Proof

Let h, k be small. Split the increment into two axis moves:

$$f(a + h, b + k) - f(a, b) = [f(a + h, b + k) - f(a, b + k)] + [f(a, b + k) - f(a, b)].$$

By the one-variable mean value theorem, there are ξ between a and $a + h$ and η between b and $b + k$ such that

$$f(a + h, b + k) - f(a, b + k) = f_x(\xi, b + k)h,$$

$$f(a, b + k) - f(a, b) = f_y(a, \eta)k.$$

Subtract the proposed linear part:

$$R = [f_x(\xi, b + k) - f_x(a, b)]h + [f_y(a, \eta) - f_y(a, b)]k.$$

By continuity, the bracketed differences tend to zero. Also $|h|, |k| \leq \sqrt{h^2 + k^2}$. Hence

$$\frac{|R|}{\sqrt{h^2 + k^2}} \rightarrow 0.$$

That is exactly differentiability.

4.3 Differentials and error propagation

The notation

$$df = f_x dx + f_y dy$$

means the derivative applied to a small displacement. It gives the first-order change

$$\Delta f \approx df.$$

For measurement errors dx, dy , this gives a fast estimate of output error.

Worked example

For $f(x, y) = \sqrt{x^2 + y^2}$,

$$df = \frac{x}{\sqrt{x^2 + y^2}} dx + \frac{y}{\sqrt{x^2 + y^2}} dy.$$

At $(3, 4)$, a change $(dx, dy) = (0.02, -0.01)$ produces

$$df \approx \frac{3}{5}(0.02) + \frac{4}{5}(-0.01) = 0.004.$$

Gradient and Directional Derivatives

5.1 The gradient

For $f : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$\nabla f = (f_x, f_y).$$

The derivative can be written as a dot product:

$$Df(\mathbf{a})(\mathbf{h}) = \nabla f(\mathbf{a}) \cdot \mathbf{h}.$$

5.2 Directional derivatives

For a unit vector $\mathbf{u}$,

$$D_{\mathbf{u}}f(\mathbf{a}) = \lim_{t \rightarrow 0} \frac{f(\mathbf{a} + t\mathbf{u}) - f(\mathbf{a})}{t}.$$

If f is differentiable,

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}.$$

Proof

Differentiability gives

$$f(\mathbf{a} + t\mathbf{u}) = f(\mathbf{a}) + Df(\mathbf{a})(t\mathbf{u}) + o(|t|).$$

Since Df is linear,

$$Df(\mathbf{a})(t\mathbf{u}) = tDf(\mathbf{a})(\mathbf{u}) = t\nabla f \cdot \mathbf{u}.$$

Divide by t and take the limit.

By Cauchy-Schwarz,

$$D_{\mathbf{u}}f = \|\nabla f\| \cos \theta.$$

Therefore the maximum directional derivative is $\|\nabla f\|$ and occurs in the gradient direction.

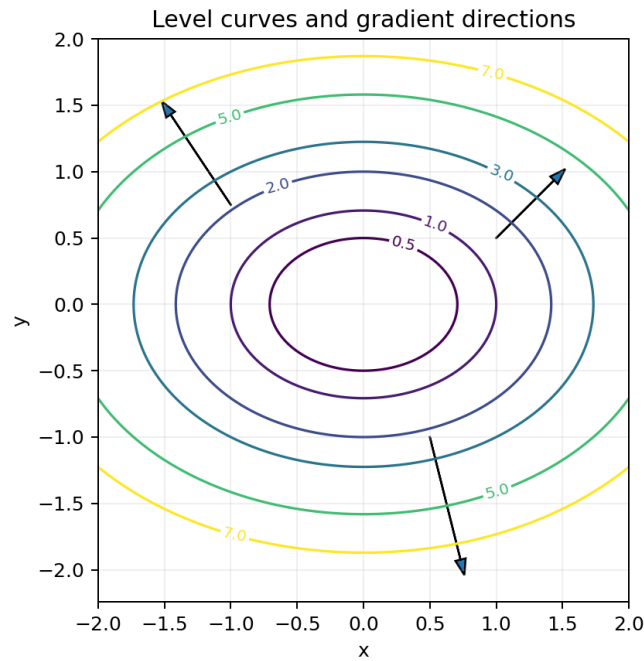


Figure 5.1: The gradient points perpendicular to level curves and toward fastest increase.

5.3 Why the gradient is normal to level curves

Suppose a differentiable curve $\mathbf{r}(t)$ lies on the level set $f = c$. Then

$$f(\mathbf{r}(t)) = c.$$

Differentiate:

$$\nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0.$$

Thus the gradient is orthogonal to every tangent vector of the level curve.

Directional derivative is a projection of the gradient

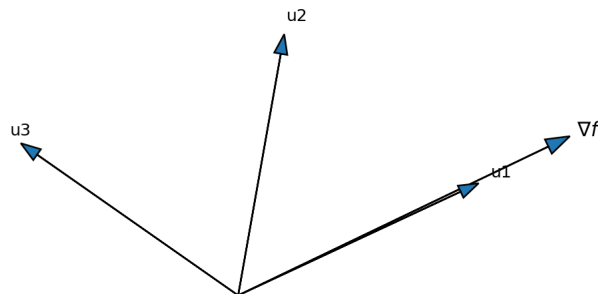


Figure 5.2: Directional change is the projection of the gradient onto the chosen direction.

Exercises

1. For $f = x^2 + 2y^2$, compute $\nabla f(1, -1)$ and the directional derivative toward $(4, 3)$.
2. Find the direction of steepest descent of $f = e^{x-y} + x^2$ at $(0, 0)$.
3. Prove directly from the dot product that $|D_{\mathbf{u}}f| \leq \|\nabla f\|$.

The Multivariable Chain Rule

6.1 One intermediate variable

Suppose $z = f(x, y)$ and $x = x(t)$, $y = y(t)$. Then z depends on t through both coordinates:

$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}.$$

Geometrically,

$$\frac{dz}{dt} = \nabla f \cdot \mathbf{r}'(t).$$

Proof

Differentiability says

$$\Delta z = f_x \Delta x + f_y \Delta y + o\left(\sqrt{(\Delta x)^2 + (\Delta y)^2}\right).$$

Divide by Δt . If x, y are differentiable, $\Delta x/\Delta t$ and $\Delta y/\Delta t$ converge, and the remainder divided by Δt tends to zero because the spatial displacement is $O(|\Delta t|)$. Taking the limit gives the formula.

6.2 Several intermediate variables and Jacobians

If $\mathbf{y} = g(\mathbf{x})$ and $\mathbf{z} = f(\mathbf{y})$, then

$$D(f \circ g)(\mathbf{x}) = Df(g(\mathbf{x})) Dg(\mathbf{x}).$$

Matrices appear because derivatives of vector-valued functions are linear maps.

For $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$,

$$F(x, y) = (u(x, y), v(x, y)),$$

the Jacobian matrix is

$$J_F = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}.$$

6.3 Tree method

If $z = f(x, y)$, $x = x(s, t)$, and $y = y(s, t)$, then

$$\frac{\partial z}{\partial s} = f_x x_s + f_y y_s, \quad \frac{\partial z}{\partial t} = f_x x_t + f_y y_t.$$

Every path from an input variable to the output contributes a product of derivatives; then add the paths.

Worked example

Let $z = e^{xy}$, $x = r \cos \theta$, $y = r \sin \theta$. Then

$$z_r = e^{xy}(yx_r + xy_r) = e^{r^2 \sin \theta \cos \theta}(y \cos \theta + x \sin \theta).$$

Substituting x, y gives

$$z_r = 2r \sin \theta \cos \theta e^{r^2 \sin \theta \cos \theta}.$$

Implicit Differentiation and the Implicit Function Theorem

7.1 Implicit relations

Not every curve is naturally given as $y = f(x)$. The circle is

$$x^2 + y^2 = 1.$$

Globally, it is not one function of x , but locally away from the top and bottom turning points, each branch is.

Suppose

$$F(x, y) = 0$$

defines y locally as a function of x . Differentiate:

$$F_x + F_y \frac{dy}{dx} = 0.$$

Therefore, when $F_y \neq 0$,

$$\boxed{\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

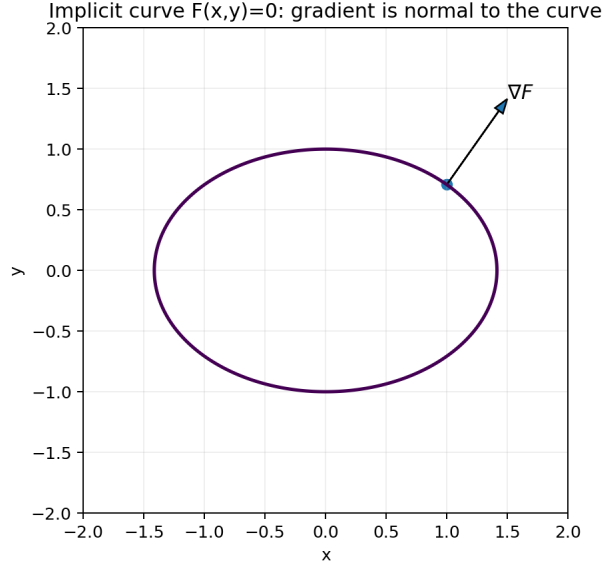


Figure 7.1: For $F(x, y) = 0$, the gradient is normal to the curve; the tangent slope follows from orthogonality.

7.2 Why the denominator matters

At a point where $F_y \neq 0$, changing y changes F to first order, so y can compensate for a small change in x to keep $F = 0$. When $F_y = 0$, that local solvability may fail or the curve may have a vertical tangent.

Theorem 7.1 (Implicit function theorem in the plane, local form). *Assume F has continuous first partial derivatives near (a, b) , $F(a, b) = 0$, and $F_y(a, b) \neq 0$. Then in some neighborhood of a there is a unique continuously differentiable function $y = g(x)$ with $g(a) = b$ and*

$$F(x, g(x)) = 0.$$

Moreover,

$$g'(x) = -\frac{F_x(x, g(x))}{F_y(x, g(x))}.$$

Proof

Because $F_y(a, b) \neq 0$ and F_y is continuous, after shrinking the rectangle around (a, b) we may assume F_y keeps one sign; suppose $F_y > 0$. For each fixed nearby x , the function $y \mapsto F(x, y)$ is then strictly increasing by the one-variable mean value theorem.

Choose $b_- < b < b_+$ close enough that $F(a, b_-) < 0 < F(a, b_+)$. Continuity in x preserves these inequalities for all x sufficiently near a . The intermediate value theorem therefore gives at least one y between b_- and b_+ with $F(x, y) = 0$, while strict monotonicity gives uniqueness. Call it $g(x)$.

To prove differentiability, compare $F(x + h, g(x + h)) = 0$ and $F(x, g(x)) = 0$. Apply the two-variable mean value decomposition along an axis path:

$$0 = F_x(\xi, g(x + h))h + F_y(x, \eta)[g(x + h) - g(x)]$$

for intermediate points ξ, η . Divide by h :

$$\frac{g(x + h) - g(x)}{h} = -\frac{F_x(\xi, g(x + h))}{F_y(x, \eta)}.$$

As $h \rightarrow 0$, continuity of g follows from the monotonic construction, so the intermediate points approach $(x, g(x))$. Continuity of F_x, F_y gives the derivative formula. Continuity of g' follows from the same formula.

7.3 Second implicit derivative

Starting from

$$F_x + F_y y' = 0,$$

differentiate with respect to x :

$$F_{xx} + 2F_{xy}y' + F_{yy}(y')^2 + F_y y'' = 0.$$

Hence

$$y'' = -\frac{F_{xx} + 2F_{xy}y' + F_{yy}(y')^2}{F_y}.$$

Exercises

1. For $x^2 + xy + y^2 = 7$, find y' at $(1, 2)$.
2. For $e^{xy} + x - y = 1$, derive dy/dx implicitly.
3. On the circle $x^2 + y^2 = 4$, identify where the implicit theorem solves locally for y as a function of x and where it fails.

Second Derivatives, Hessians, and Quadratic Approximation

8.1 The Hessian

For a twice differentiable scalar function,

$$H_f = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}.$$

When mixed partials agree, this matrix is symmetric.

The gradient describes slope. The Hessian describes how the slope itself changes: curvature.

8.2 Two-variable Taylor polynomial

Near (a, b) , let $h = x - a$, $k = y - b$. The second-order Taylor approximation is

$$\begin{aligned} f(a + h, b + k) \approx & f(a, b) + f_x h + f_y k \\ & + \frac{1}{2} (f_{xx} h^2 + 2f_{xy} h k + f_{yy} k^2). \end{aligned}$$

In matrix form,

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a})^T \mathbf{h} + \frac{1}{2} \mathbf{h}^T H_f(\mathbf{a}) \mathbf{h}.$$

8.3 Proof from one-variable Taylor theory

Theorem 8.1 (Taylor formula along a line). *Assume f has continuous derivatives through order $m + 1$ near $\mathbf{a}$. Define*

$$g(t) = f(\mathbf{a} + t\mathbf{h}).$$

Then the multivariable Taylor expansion is obtained by applying the ordinary one-variable Taylor theorem to g at $t = 0$ and setting $t = 1$.

Proof

By the chain rule,

$$g'(0) = \nabla f(\mathbf{a}) \cdot \mathbf{h}.$$

Differentiating again,

$$g''(0) = \mathbf{h}^T H_f(\mathbf{a}) \mathbf{h}.$$

The one-variable Taylor theorem with Lagrange remainder gives

$$g(1) = g(0) + g'(0) + \frac{1}{2}g''(0) + \frac{g^{(3)}(\xi)}{3!}$$

for some $\xi \in (0, 1)$ in the third-order case. Substituting the expressions for $g(0)$, $g'(0)$, $g''(0)$ yields the two-variable formula. Higher orders follow by repeated directional differentiation.

Function and second-order Taylor approximation near (0,0)

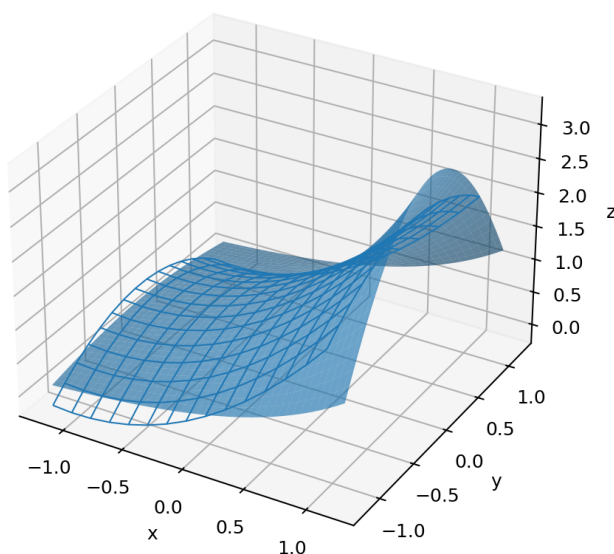


Figure 8.1: $e^x \cos y$ and its second-order Taylor approximation near the origin.

8.4 Example: Taylor expansion of $e^x \cos y$

At $(0, 0)$,

$$\begin{aligned} f &= 1, & f_x &= 1, & f_y &= 0, \\ f_{xx} &= 1, & f_{xy} &= 0, & f_{yy} &= -1. \end{aligned}$$

Therefore

$$e^x \cos y = 1 + x + \frac{x^2 - y^2}{2} + O(\|(x, y)\|^3).$$

8.5 Higher-order two-variable series

If f is analytic, formally

$$f(a + h, b + k) = \sum_{m, n \geq 0} \frac{1}{m!n!} \frac{\partial^{m+n} f}{\partial x^m \partial y^n}(a, b) h^m k^n.$$

For actual convergence to f , analyticity is required; infinite differentiability by itself does not guarantee equality with the Taylor series.

Critical Points, Quadratic Forms, and Optimization

9.1 Critical points

For an interior local maximum or minimum of a differentiable function,

$$\nabla f = 0.$$

Why? Every one-dimensional slice through the point has a local extremum, so every directional derivative is zero; hence the gradient is zero.

9.2 Second derivative test

At a critical point, the linear term vanishes. The first meaningful local shape is the quadratic form

$$Q(h, k) = f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2.$$

Let

$$D = f_{xx}f_{yy} - f_{xy}^2 = \det H_f.$$

Then:

- if $D > 0$ and $f_{xx} > 0$, local minimum;
- if $D > 0$ and $f_{xx} < 0$, local maximum;
- if $D < 0$, saddle point;
- if $D = 0$, the test is inconclusive.

Proof

For $Q = Ah^2 + 2Bhk + Ck^2$, complete the square when $A \neq 0$:

$$Q = A \left(h + \frac{B}{A}k \right)^2 + \frac{AC - B^2}{A}k^2.$$

If $AC - B^2 > 0$, both coefficients have the sign of A , so Q has one sign for every nonzero (h, k) . If $AC - B^2 < 0$, choose directions that make the two terms compete; Q takes both signs. Taylor's theorem says the actual function differs from $f(a, b) + \frac{1}{2}Q$ by a smaller-order term, so sufficiently close to the critical point the sign of Q determines the local shape.

9.3 Lagrange multipliers

To optimize $f(x, y)$ subject to $g(x, y) = c$, a constrained extremum typically satisfies

$$\nabla f = \lambda \nabla g.$$

Proof

At a constrained extremum, any allowed infinitesimal motion is tangent to the constraint curve. If $\mathbf{t}$ is a tangent vector, then $\nabla g \cdot \mathbf{t} = 0$. Because moving along the constraint cannot change f to first order at the extremum, also $\nabla f \cdot \mathbf{t} = 0$. In the plane, two nonzero vectors perpendicular to the same one-dimensional tangent direction are parallel. Hence $\nabla f = \lambda \nabla g$.

Worked example

Maximize xy subject to $x^2 + y^2 = 1$. We have

$$\nabla f = (y, x), \quad \nabla g = (2x, 2y).$$

Thus $y = 2\lambda x$ and $x = 2\lambda y$. Nonzero solutions imply $x^2 = y^2$. With the constraint, $|x| = |y| = 1/\sqrt{2}$. Hence the maximum is $1/2$ and minimum is $-1/2$.

Exercises

1. Find and classify the critical points of $x^2 + xy + y^2 - 4x$.
2. Find the second-order Taylor polynomial of $\log(1 + x + y)$ at $(0, 0)$.
3. Maximize $x + 2y$ on the ellipse $x^2 + 4y^2 = 4$ using Lagrange multipliers.
4. Show that $f = x^4 + y^4$ has a strict minimum at the origin even though the second derivative test is inconclusive there.

Part III

Integration: Accumulation, Inversion, and Change of Variables

The Definite Integral from First Principles

10.1 The real meaning: accumulation

A derivative asks: *what is the instantaneous rate?* An integral asks the inverse-style question: *if tiny contributions are added, what total quantity results?*

Suppose a density $f(x)$ gives quantity per unit length. Split $[a, b]$ into small pieces of widths Δx_i . On a tiny piece, f is approximately constant, so the contribution is approximately

$$f(x_i^*)\Delta x_i.$$

Adding gives a Riemann sum

$$\sum_i f(x_i^*)\Delta x_i.$$

The definite integral is the limit as the mesh becomes arbitrarily fine.

Definition 10.1 (Riemann integral). Let $P : a = x_0 < x_1 < \cdots < x_n = b$ be a partition, $\Delta x_i = x_i - x_{i-1}$, and choose $x_i^* \in [x_{i-1}, x_i]$. If there is a number I such that every sufficiently fine tagged partition has

$$\sum_{i=1}^n f(x_i^*)\Delta x_i \rightarrow I,$$

then

$$I = \int_a^b f(x) dx.$$

The symbol dx reminds us that the tiny contribution is “height times a tiny x -width.” The integral is signed: negative values of f contribute negative accumulation.

10.2 Area is an interpretation, not the definition

When $f \geq 0$, the integral equals geometric area under the graph. But the same definition computes mass from linear density, distance from velocity, charge from charge density, work from variable force, probability from density, and many other totals. Area is the easiest picture of a more general accumulation operation.

10.3 The Fundamental Theorem of Calculus

Define the accumulation function

$$F(x) = \int_a^x f(t) dt.$$

If f is continuous, then

$$F'(x) = f(x).$$

This is the deepest reason differentiation and integration are inverse operations.

Proof

Consider

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) dt.$$

Because f is continuous at x , for every $\varepsilon > 0$ there exists δ such that $|f(t) - f(x)| < \varepsilon$ whenever $|t - x| < \delta$. For $|h| < \delta$,

$$\left| \frac{1}{h} \int_x^{x+h} f(t) dt - f(x) \right| = \left| \frac{1}{h} \int_x^{x+h} (f(t) - f(x)) dt \right| \leq \frac{1}{|h|} \int_x^{x+h} \varepsilon |dt| = \varepsilon.$$

Therefore the difference quotient tends to $f(x)$.

The second half follows immediately: if $G' = f$, then $G - F$ has derivative zero, so it is constant. Thus

$$\int_a^b f(x) dx = G(b) - G(a).$$

10.4 Why antiderivatives calculate integrals

The definite integral is defined by a limit of sums, not by antiderivatives. The Fundamental Theorem is what allows us to replace an impossible-looking limit of infinitely many contributions by endpoint evaluation.

Worked example

Compute $\int_0^2 (3x^2 + 1) dx$. An antiderivative is $G = x^3 + x$, so

$$\int_0^2 (3x^2 + 1) dx = (8 + 2) - (0) = 10.$$

The number 10 is the net accumulated contribution, not merely “the area because integrals mean area.”

Substitution: Rewriting Accumulation in a Better Coordinate

11.1 The idea

Suppose an integral contains a complicated expression $g(x)$ whose derivative also appears. Let

$$u = g(x).$$

Then an x -increment dx produces approximately

$$du = g'(x)dx.$$

So

$$\int f(g(x))g'(x) dx$$

should become

$$\int f(u) du.$$

This is the chain rule run backward.

Theorem 11.1 (Substitution rule). *If g is continuously differentiable and f is continuous, then*

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

whenever the ordinary substitution hypotheses are satisfied.

Proof

Let F be an antiderivative of f . By the chain rule,

$$\frac{d}{dx}F(g(x)) = F'(g(x))g'(x) = f(g(x))g'(x).$$

The Fundamental Theorem gives

$$\int_a^b f(g(x))g'(x) dx = F(g(b)) - F(g(a)).$$

But the Fundamental Theorem in the u variable gives

$$\int_{g(a)}^{g(b)} f(u) du = F(g(b)) - F(g(a)).$$

Hence the two integrals are equal.

Method

When should you try substitution?

1. Identify an “inside function” $g(x)$.
2. Check whether $g'(x)$ appears, up to a constant factor.
3. Set $u = g(x)$ and replace every occurrence consistently.
4. For definite integrals, either change the bounds immediately or return to x before evaluating. Do not mix variables and bounds.

Worked example

$$\int x e^{x^2} dx.$$

Set $u = x^2$, so $du = 2x dx$. Then

$$\int x e^{x^2} dx = \frac{1}{2} \int e^u du = \frac{1}{2} e^u + C = \boxed{\frac{1}{2} e^{x^2} + C}.$$

Integration by Parts: Reversing the Product Rule

The product rule is

$$(uv)' = u'v + uv'.$$

Rearrange:

$$uv' = (uv)' - u'v.$$

Integrate:

$$\boxed{\int u \, dv = uv - \int v \, du.}$$

So integration by parts is exactly the product rule run backward.

Proof

Integrate the product-rule identity from a to b :

$$\int_a^b (uv)' \, dx = \int_a^b u'v \, dx + \int_a^b uv' \, dx.$$

The Fundamental Theorem gives

$$[u(x)v(x)]_a^b = \int_a^b u'v \, dx + \int_a^b uv' \, dx.$$

Rearrange.

12.1 Choosing u

A useful heuristic is to choose u so differentiation simplifies it, while dv is easy to integrate. Common priority: logarithms, inverse trig functions, algebraic polynomials, trig, exponentials. This is a heuristic, not a theorem.

Worked example

Compute $\int xe^x \, dx$. Choose $u = x$, $dv = e^x dx$. Then $du = dx$, $v = e^x$:

$$\int xe^x \, dx = xe^x - \int e^x \, dx = \boxed{e^x(x - 1) + C}.$$

Worked example

Compute $\int \log x \, dx$. Think of $\log x$ as $(\log x) \cdot 1$. Let $u = \log x$, $dv = dx$, so $du = dx/x$, $v = x$:

$$\int \log x \, dx = x \log x - \int 1 \, dx = \boxed{x \log x - x + C}.$$

12.2 Substitution versus integration by parts

Substitution recognizes a hidden chain rule. Integration by parts recognizes a hidden product rule. When deciding between them, ask which differentiation rule would naturally produce the integrand.

Exercises

1. Evaluate $\int_0^1 2x(1+x^2)^5 \, dx$ by substitution.
2. Evaluate $\int x \cos x \, dx$ by parts.
3. Evaluate $\int x^2 e^x \, dx$ by repeated parts.
4. Evaluate $\int_0^{\pi/2} \sin^3 x \cos x \, dx$.
5. Explain in one sentence why substitution is the inverse of the chain rule and integration by parts is the inverse of the product rule.

Double Integrals: Accumulation over Area

13.1 Definition from two-dimensional Riemann sums

Let $R = [a, b] \times [c, d]$. Split it into small rectangles of area $\Delta A_{ij} = \Delta x_i \Delta y_j$. Choose a point (x_{ij}^*, y_{ij}^*) in each cell. Then

$$\sum_{i,j} f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}$$

approximates the total accumulation. The limit is

$$\iint_R f(x, y) dA.$$

If f is a surface height and $f \geq 0$, this is volume. If f is surface density, it is mass. If f is probability density, it is probability.

13.2 Iterated integrals and Fubini's theorem

For continuous f on a rectangle,

$$\iint_R f dA = \int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_c^d \left(\int_a^b f(x, y) dx \right) dy.$$

Proof

For a rectangular grid, the finite Riemann sum can be grouped by columns:

$$\sum_i \sum_j f(x_i^*, y_j^*) \Delta y_j \Delta x_i.$$

For each fixed i , as the y -mesh shrinks the inner sum tends to $\int_c^d f(x_i^*, y) dy$. Then the outer sum tends to the integral in x . Continuity gives uniform control needed to pass the limits. Grouping by rows gives the reverse order. Thus both iterated integrals equal the same two-dimensional Riemann integral.

13.3 Nonrectangular regions

If

$$R = \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\},$$

then

$$\iint_R f \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx.$$

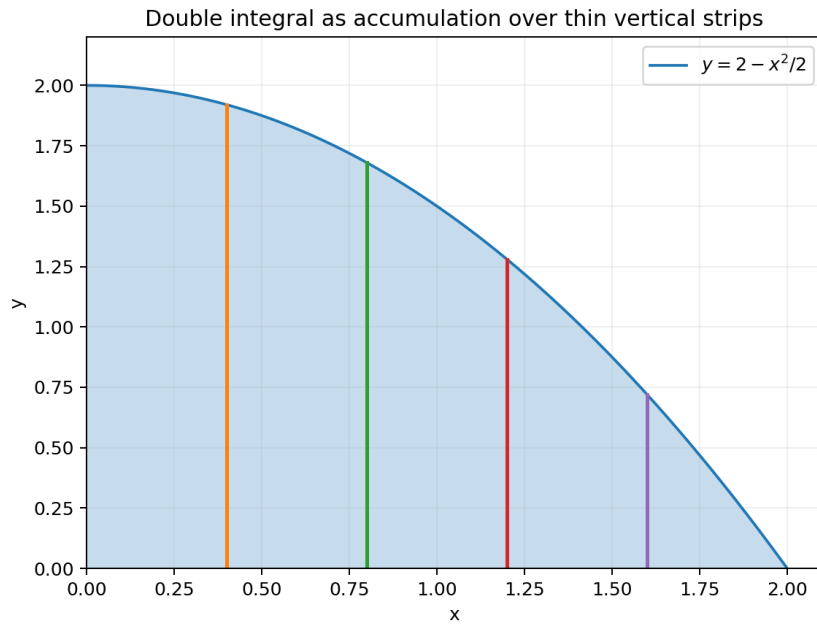


Figure 13.1: The outer integral adds the contributions of thin vertical strips.

Worked example

Find the area of the triangle $R = \{x \geq 0, y \geq 0, x + y \leq 1\}$. Use $f = 1$:

$$\iint_R 1 \, dA = \int_0^1 \int_0^{1-x} 1 \, dy \, dx = \int_0^1 (1 - x) \, dx = \frac{1}{2}.$$

13.4 Changing the order

Changing order is geometric, not merely algebraic. Draw the region, project it onto the new outer axis, and determine the corresponding inner bounds. The function stays the same; only the slicing changes.

Change of Variables and the Jacobian

14.1 Why a Jacobian factor is necessary

In one-dimensional substitution, lengths stretch by approximately $|g'(u)|$. In two dimensions, a coordinate map

$$T(u, v) = (x(u, v), y(u, v))$$

turns a tiny $du dv$ rectangle into a small parallelogram. The area scaling is the absolute determinant of the derivative matrix:

$$J = \det \frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}.$$

Therefore

$$dA = |J| du dv.$$

Proof

At a point, differentiability gives

$$T(\mathbf{u} + \Delta \mathbf{u}) \approx T(\mathbf{u}) + DT(\mathbf{u})\Delta \mathbf{u}.$$

So to first order, the coordinate transformation acts like the linear map DT . A tiny rectangle spanned by $(du, 0)$ and $(0, dv)$ is mapped to the parallelogram spanned by the columns of DT multiplied by du and dv . The area of that parallelogram is $|\det DT| du dv$. Summing tiny cells and taking the limit gives the change-of-variables formula under standard smooth one-to-one hypotheses.

14.2 Polar coordinates

Set

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Then

$$J = \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} = r.$$

Hence

$$dA = r dr d\theta.$$

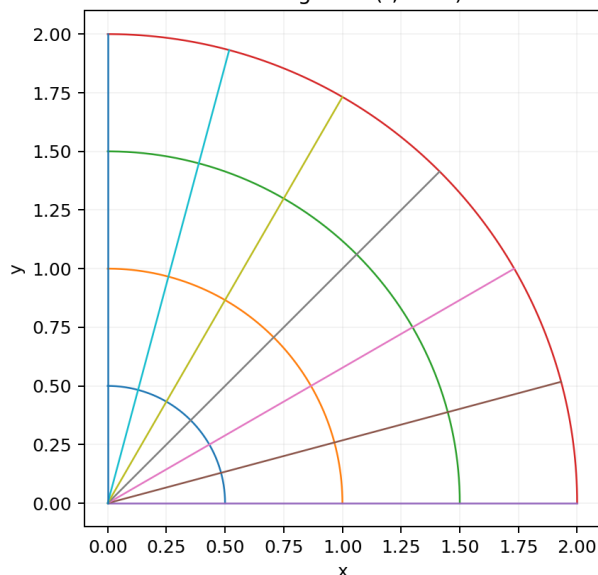
Polar substitution: small rectangles in (r, θ) become curved sectors

Figure 14.1: The factor r appears because angular separation corresponds to a physical arc length $r d\theta$.

Worked example

Compute the area of a disk of radius R :

$$\int_0^{2\pi} \int_0^R r \, dr \, d\theta = \int_0^{2\pi} \frac{R^2}{2} \, d\theta = \pi R^2.$$

This derivation reveals where the familiar formula comes from.

Exercises

1. Compute $\iint_{[0,1]^2} (x+y) \, dA$ in both orders.
2. Reverse the order of $\int_0^1 \int_x^1 f(x,y) \, dy \, dx$.
3. Use polar coordinates to compute $\iint_{x^2+y^2 \leq 1} (x^2+y^2) \, dA$.
4. Find the Jacobian of $x = u+v$, $y = u-v$ and explain its area scaling.
5. Compute the mass of the unit disk with density $\rho(x,y) = x^2 + y^2$.

Part IV

Differential Equations: From Local Laws to Global Motion

What a Differential Equation Really Says

15.1 The central idea

An algebraic equation constrains values. A differential equation constrains *change*. For example,

$$y' = 3y$$

says that at every instant the rate of change is three times the current amount. The solution is not a number but a function.

15.2 Order and linearity

The order is the highest derivative. A first-order ODE has the form

$$F(x, y, y') = 0.$$

A second-order ODE has the form

$$F(x, y, y', y'') = 0.$$

A linear second-order equation is

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = g(x).$$

Nonlinear examples include $y' = y^2 - x$ and $y'' + \sin y = 0$.

15.3 Explicit versus implicit solutions

An explicit solution solves directly for the dependent variable:

$$y(x) = Ce^x.$$

An implicit solution is a relation

$$F(x, y) = C$$

that solution curves satisfy. It may be impossible or undesirable to isolate y .

Worked example

The equation

$$y' = \frac{x}{y}$$

gives $y \, dy = x \, dx$, hence

$$\boxed{y^2 - x^2 = C.}$$

This is already a valid implicit general solution. Solving for y gives branches

$$y = \pm \sqrt{x^2 + C},$$

but branch choices and domains must be handled separately.

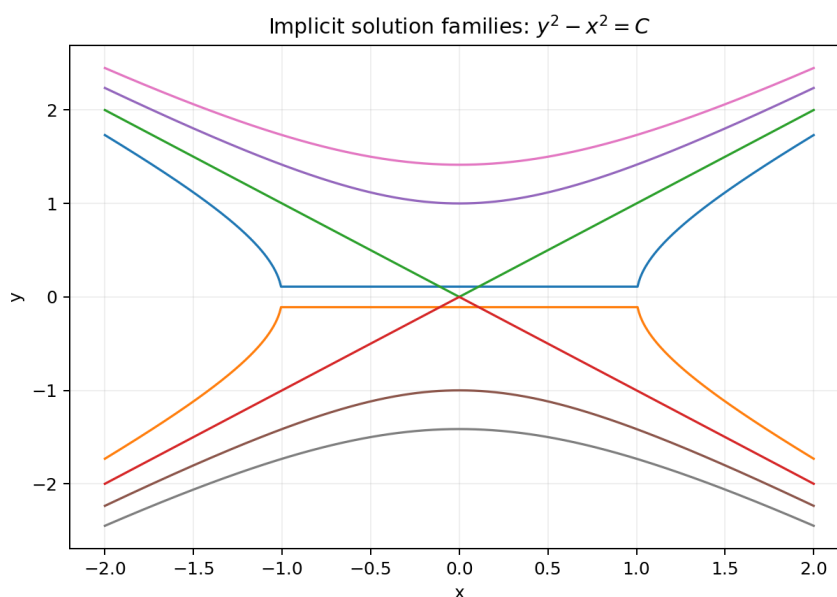


Figure 15.1: An implicit ODE solution can naturally describe an entire family of solution curves.

15.4 Initial-value problems

An n th-order ODE typically needs n independent initial conditions. For example,

$$y'' + y = 0, \quad y(0) = 2, \quad y'(0) = 1.$$

The differential equation gives a family; initial data selects one member.

15.5 Direction fields

For

$$y' = f(x, y),$$

every point (x, y) is assigned a slope $f(x, y)$. A solution curve is a curve whose tangent obeys that slope rule at every point.

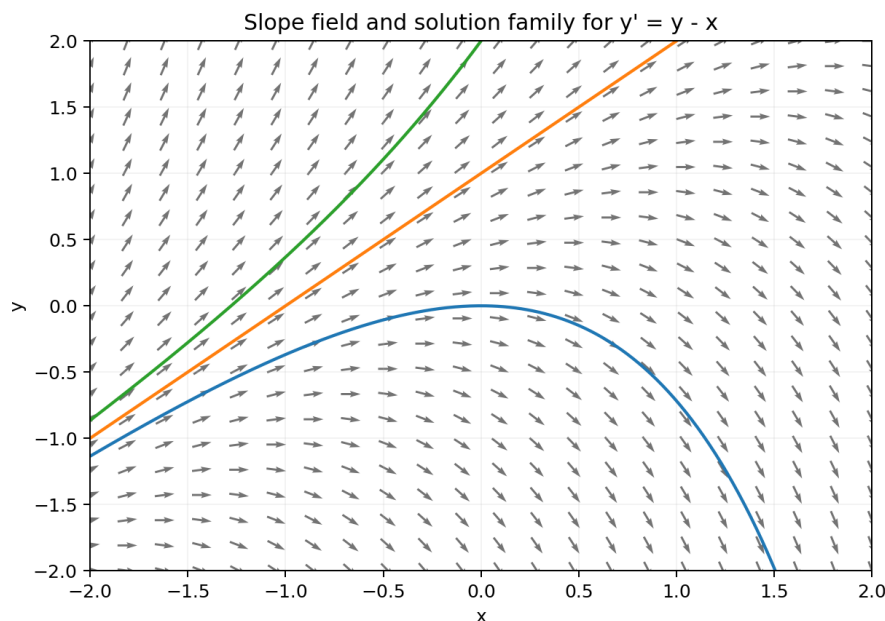


Figure 15.2: The ODE is a field of local instructions; a solution follows those instructions.

15.6 No universal closed-form algorithm

There is no formula that solves every first- or second-order ODE in elementary functions. Even simple-looking nonlinear equations can require special functions, implicit descriptions, series, qualitative analysis, or numerical approximation. Mastery therefore means two things:

1. recognize important solvable structures;
2. know what to do when no symbolic structure applies.

First-Order ODEs: A Recognition and Solution Toolkit

16.1 Separable equations

If

$$y' = g(x)h(y),$$

then, away from zeros of h ,

$$\frac{dy}{h(y)} = g(x)dx.$$

Integrate both sides.

Proof

Let $H'(y) = 1/h(y)$. By the chain rule,

$$\frac{d}{dx}H(y(x)) = H'(y)y' = \frac{1}{h(y)}g(x)h(y) = g(x).$$

Integrating gives $H(y) = \int g(x)dx + C$. Thus separation is shorthand for a chain-rule argument, not literal algebra with infinitesimals.

Common trap

Before dividing by $h(y)$, check equilibrium solutions $h(y) = 0$. Dividing can silently lose them.

Worked example

Solve $y' = xy$, $y(0) = 3$.

$$\frac{dy}{y} = x dx, \quad \ln |y| = \frac{x^2}{2} + C.$$

Thus $y = Ce^{x^2/2}$. The initial condition gives $C = 3$, so

$$y = 3e^{x^2/2}.$$

16.2 Autonomous equations and equilibria

An autonomous equation is

$$y' = f(y).$$

Equilibria satisfy $f(y_*) = 0$. Sign analysis on intervals tells whether solutions increase or decrease. If $f'(y_*) < 0$, nearby perturbations tend to decay; if $f'(y_*) > 0$, they tend to grow.

16.3 First-order linear equations

The standard form is

$$y' + p(x)y = q(x).$$

We seek a factor μ making the left side a product derivative:

$$(\mu y)' = \mu y' + \mu' y.$$

We need $\mu' = p\mu$, so

$$\mu = e^{\int p(x)dx}.$$

Then

$$(\mu y)' = \mu q,$$

and

$$y = \frac{1}{\mu} \left(\int \mu q dx + C \right).$$

Worked example

Solve $y' + 2y = e^x$.

$$\mu = e^{2x}, \quad (e^{2x}y)' = e^{3x}.$$

Thus

$$e^{2x}y = \frac{1}{3}e^{3x} + C,$$

so

$$y = \frac{1}{3}e^x + Ce^{-2x}.$$

16.4 Exact equations

An equation

$$M(x, y)dx + N(x, y)dy = 0$$

is exact if there is a potential F with

$$F_x = M, \quad F_y = N.$$

Then

$$dF = 0 \quad \Rightarrow \quad F(x, y) = C.$$

If M, N have continuous first derivatives on a simply connected region, exactness is characterized by

$$M_y = N_x.$$

Worked example

Solve

$$(2xy + 3)dx + (x^2 + 4y)dy = 0.$$

Since $M_y = 2x = N_x$, it is exact. Integrate $F_x = 2xy + 3$ in x :

$$F = x^2y + 3x + g(y).$$

Then $F_y = x^2 + g'(y) = x^2 + 4y$, so $g = 2y^2$. Hence

$$x^2y + 3x + 2y^2 = C.$$

This is naturally an implicit solution.

16.5 Bernoulli equations

For

$$y' + p(x)y = q(x)y^n,$$

with $n \neq 0, 1$, divide by y^n and set

$$v = y^{1-n}.$$

Then

$$v' = (1-n)y^{-n}y',$$

which turns the ODE into a first-order linear equation in v .

16.6 Homogeneous first-order equations

If

$$y' = F(y/x),$$

use

$$v = y/x, \quad y = vx, \quad y' = v + xv'.$$

The resulting equation is often separable.

16.7 Recognition tree

Given a first-order ODE, ask in order:

1. Is it directly separable?
2. Can it be rearranged into $y' + p(x)y = q(x)$?
3. In differential form $Mdx + Ndy = 0$, is it exact?
4. Is it Bernoulli?
5. Does it depend on y/x or another standard homogeneous substitution?
6. Is there a useful substitution suggested by the algebra?
7. If none applies, can qualitative analysis, series, or numerical methods answer the question?

Existence, Uniqueness, and Why Initial Data Usually Selects One Curve

17.1 Integral formulation

The IVP

$$y' = f(x, y), \quad y(x_0) = y_0$$

is equivalent to

$$y(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt.$$

This converts the differential problem into a fixed-point problem.

Theorem 17.1 (Picard-Lindelof, local form). *If f is continuous near (x_0, y_0) and locally Lipschitz in y , meaning*

$$|f(x, u) - f(x, v)| \leq L|u - v|,$$

then the IVP has a unique local solution.

17.2 Picard iteration

Start with $y_0(x) = y_0$ and define

$$y_{n+1}(x) = y_0 + \int_{x_0}^x f(t, y_n(t)) dt.$$

Under a sufficiently small interval and a Lipschitz condition, this is a contraction and converges uniformly to the solution.

Proof

Let $T[y](x) = y_0 + \int_{x_0}^x f(t, y(t)) dt$. On an interval of half-width h ,

$$\|T[u] - T[v]\|_\infty \leq \sup_x \int_{x_0}^x L|u(t) - v(t)| dt \leq Lh \|u - v\|_\infty.$$

Choose h so $Lh < 1$. Then T is a contraction on a suitable closed set of continuous functions. The contraction mapping theorem gives a unique fixed point $y = T[y]$, which is exactly the integral equation and therefore the ODE solution.

17.3 A direct uniqueness estimate

If y_1, y_2 are two solutions with the same initial value,

$$|y_1(x) - y_2(x)| \leq L \int_{x_0}^x |y_1(t) - y_2(t)| dt.$$

Grönwall's inequality implies the difference is zero.

17.4 How uniqueness can fail

Consider

$$y' = \sqrt{|y|}, \quad y(0) = 0.$$

One solution is $y = 0$. For any $a \geq 0$, another is

$$y(x) = \begin{cases} 0, & x \leq a, \\ (x - a)^2/4, & x > a. \end{cases}$$

The right side is continuous but not Lipschitz in y near zero. Thus existence may hold while uniqueness fails.

Second-Order ODEs: Structure Before Technique

18.1 Linear equations

A general linear second-order ODE is

$$a(x)y'' + b(x)y' + c(x)y = g(x).$$

The homogeneous equation has $g = 0$. If y_1, y_2 solve the homogeneous equation, then so does every linear combination $C_1y_1 + C_2y_2$.

Proof

Let $L[y] = ay'' + by' + cy$. Linearity gives

$$L[C_1y_1 + C_2y_2] = C_1L[y_1] + C_2L[y_2] = 0.$$

A general nonhomogeneous solution is

$$y = y_h + y_p,$$

where $L[y_h] = 0$ and $L[y_p] = g$.

18.2 Constant coefficients and the characteristic equation

For

$$ay'' + by' + cy = 0,$$

try $y = e^{rx}$. Then

$$(ar^2 + br + c)e^{rx} = 0.$$

Because $e^{rx} \neq 0$,

$$ar^2 + br + c = 0.$$

18.2.1 Distinct real roots

If $r_1 \neq r_2$,

$$y = C_1e^{r_1x} + C_2e^{r_2x}.$$

18.2.2 Repeated root

If r is repeated,

$$y = (C_1 + C_2x)e^{rx}.$$

A useful derivation is to let two distinct roots coalesce:

$$\lim_{\varepsilon \rightarrow 0} \frac{e^{(r+\varepsilon)x} - e^{rx}}{\varepsilon} = xe^{rx}.$$

18.2.3 Complex roots

If $r = \alpha \pm i\beta$, Euler's formula gives

$$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x).$$

18.3 Wronskian and linear independence

For two solutions,

$$W = y_1 y_2' - y_2 y_1'.$$

If $W(x_0) \neq 0$, they are linearly independent. For

$$y'' + p(x)y' + q(x)y = 0,$$

Abel's identity is

$$W(x) = Ce^{-\int p(x)dx}.$$

Proof

Differentiate:

$$W' = y_1 y_2'' - y_2 y_1''.$$

Using $y_i'' = -py_i' - qy_i$,

$$W' = -p(y_1 y_2' - y_2 y_1') = -pW.$$

This first-order separable equation gives the formula.

18.4 Undetermined coefficients

For constant-coefficient equations with forcing made from polynomials, exponentials, sines, and cosines, guess a particular solution of the same algebraic family. If the guess overlaps the homogeneous solution, multiply by a sufficient power of x .

Worked example

Solve

$$y'' - 3y' + 2y = e^{4x}.$$

The homogeneous roots are 1, 2, so $y_h = C_1 e^x + C_2 e^{2x}$. Guess $y_p = A e^{4x}$. Substitution gives

$$(16 - 12 + 2)A e^{4x} = e^{4x},$$

so $A = 1/6$. Therefore

$$y = C_1 e^x + C_2 e^{2x} + \frac{1}{6} e^{4x}.$$

18.5 Variation of parameters

Suppose y_1, y_2 form a fundamental homogeneous pair. Seek

$$y_p = u_1 y_1 + u_2 y_2.$$

Impose $u'_1 y_1 + u'_2 y_2 = 0$. Then substitution gives

$$u'_1 y'_1 + u'_2 y'_2 = g$$

for the normalized equation $y'' + py' + qy = g$. Solving,

$$u'_1 = -\frac{y_2 g}{W}, \quad u'_2 = \frac{y_1 g}{W}.$$

This method is more general than undetermined coefficients.

18.6 Cauchy-Euler equations

For

$$x^2 y'' + axy' + by = 0,$$

try $y = x^m$. Then

$$[m(m-1) + am + b]x^m = 0,$$

so solve the algebraic indicial equation

$$m(m-1) + am + b = 0.$$

This is the scale-invariant analogue of the exponential ansatz for constant coefficients.

18.7 Reduction of order

If a second-order equation lacks y explicitly,

$$y'' = F(x, y'),$$

set $v = y'$ to obtain a first-order equation for v .

If it lacks x explicitly,

$$y'' = F(y, y'),$$

set $v(y) = y'$. Then by the chain rule

$$y'' = \frac{dv}{dx} = \frac{dv}{dy} \frac{dy}{dx} = v \frac{dv}{dy},$$

which may reduce the order.

Mechanical Meaning: Oscillations, Damping, and Resonance

19.1 The harmonic oscillator

Hooke's law $F = -kx$ and Newton's law $F = mx''$ give

$$mx'' + kx = 0.$$

Set $\omega_0 = \sqrt{k/m}$. Then

$$x'' + \omega_0^2 x = 0,$$

with solution

$$x = A \cos \omega_0 t + B \sin \omega_0 t.$$

19.2 Energy conservation

Multiply the ODE by x' :

$$mx''x' + kxx' = 0.$$

Recognize derivatives:

$$\frac{d}{dt} \left(\frac{1}{2} m (x')^2 + \frac{1}{2} k x^2 \right) = 0.$$

Thus total energy is constant.

19.3 Damping

With drag proportional to velocity,

$$mx'' + cx' + kx = 0.$$

The discriminant $c^2 - 4mk$ determines overdamped, critically damped, or underdamped behavior. In the underdamped case,

$$x = e^{-ct/(2m)} (A \cos \omega_d t + B \sin \omega_d t).$$

The exponential envelope is the mathematical signature of energy loss.

19.4 Forced oscillation and resonance

For

$$x'' + \omega_0^2 x = F \cos \omega t,$$

when $\omega \neq \omega_0$, a particular solution has amplitude proportional to

$$\frac{1}{|\omega_0^2 - \omega^2|}.$$

At exact resonance in the undamped model, the usual cosine guess overlaps the homogeneous solution, and a particular solution contains

$$t \sin \omega_0 t,$$

so amplitude grows linearly with time.

Euler's Method: Building a Curve from Its Local Slopes

20.1 Derivation

For

$$y' = f(t, y), \quad y(t_0) = y_0,$$

Taylor's theorem says

$$y(t + h) = y(t) + hy'(t) + \frac{h^2}{2}y''(\xi).$$

Discarding the $O(h^2)$ term gives

$$y_{n+1} = y_n + hf(t_n, y_n), \quad t_{n+1} = t_n + h.$$

Intuition

The ODE supplies the tangent slope at the current point. Euler's method follows that tangent for a short horizontal step h , then asks the ODE for a new slope, and repeats.

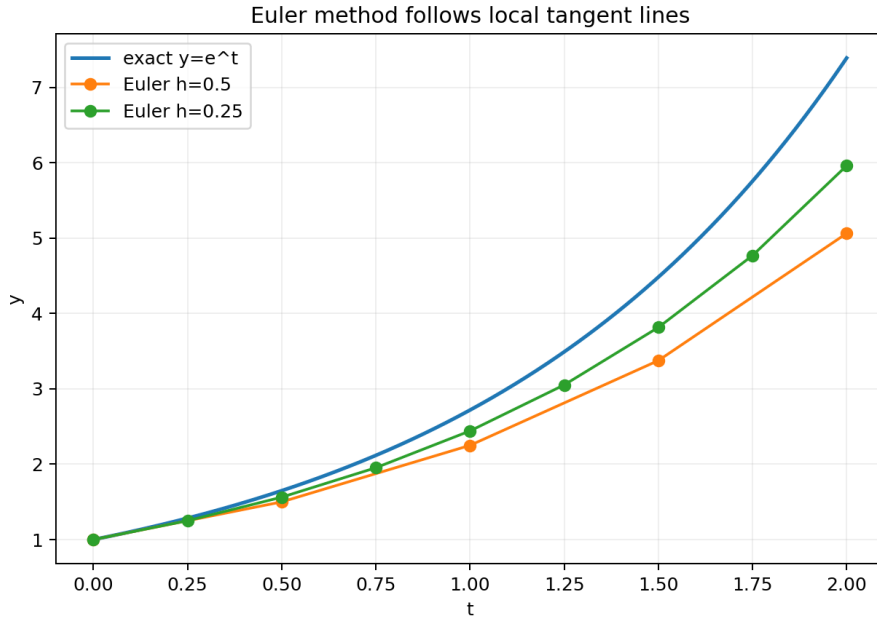


Figure 20.1: Smaller steps force the polygonal Euler path to follow the exact exponential solution more closely.

20.2 Local and global error

One Euler step drops a term of size $O(h^2)$, so the local truncation error is $O(h^2)$. Over a fixed interval there are about $1/h$ steps, producing global error $O(h)$ under standard smoothness and stability hypotheses.

Theorem 20.1 (Basic global error estimate). *Suppose f is Lipschitz in y with constant L and the exact solution has $|y''(t)| \leq M$ on the interval. Then for Euler's method with step h , the error at t_n satisfies a bound of the form*

$$|e_n| \leq \frac{Mh}{2L}(e^{L(t_n-t_0)} - 1)$$

when $L > 0$.

Proof

Taylor gives

$$y(t_{n+1}) = y(t_n) + hf(t_n, y(t_n)) + \tau_n, \quad |\tau_n| \leq \frac{M}{2}h^2.$$

Subtract the Euler update:

$$e_{n+1} = e_n + h[f(t_n, y(t_n)) - f(t_n, y_n)] + \tau_n.$$

Taking absolute values and using Lipschitz continuity,

$$|e_{n+1}| \leq (1 + hL)|e_n| + \frac{M}{2}h^2.$$

Iterating this geometric recurrence from $e_0 = 0$ gives

$$|e_n| \leq \frac{Mh}{2L}[(1 + hL)^n - 1].$$

Since $(1 + hL)^n \leq e^{Ln timer}$, the stated bound follows.

20.3 Numerical stability

For the test equation $y' = \lambda y$, Euler gives

$$y_{n+1} = (1 + h\lambda)y_n.$$

If $\lambda < 0$, the exact solution decays. Euler also decays only if

$$|1 + h\lambda| < 1.$$

Thus

$$0 < h < \frac{2}{|\lambda|}.$$

A numerical method can therefore become unstable even when the true ODE is stable.

20.4 Improved methods

Euler uses one slope. Heun's method averages a starting and predicted ending slope. Classical RK4 uses four slopes and has global error $O(h^4)$. These methods preserve the same core philosophy: infer finite motion from local derivative information.

A Practical ODE Decision Guide

First-order decision guide

Given $F(x, y, y') = 0$:

1. Solve algebraically for y' if possible.
2. Check equilibrium/singular solutions before dividing.
3. Test separability.
4. Test linear form $y' + py = q$.
5. Rewrite as $Mdx + Ndy = 0$ and test exactness.
6. Test Bernoulli or y/x -homogeneous structure.
7. Search for a simplifying substitution suggested by repeated combinations.
8. If symbolic methods fail, describe direction fields/equilibria and use Euler/RK methods.

Second-order decision guide

Given $F(x, y, y', y'') = 0$:

1. Is it linear? If yes, solve the homogeneous part first.
2. Constant coefficients: characteristic polynomial.
3. Cauchy-Euler coefficients: try $y = x^m$.
4. Known homogeneous solution: reduction of order may produce another.
5. Simple forcing with constant coefficients: undetermined coefficients.
6. General forcing with known fundamental pair: variation of parameters.
7. Missing y or missing x : reduce order using $v = y'$ or $v(y) = y'$.
8. If nonlinear without special structure, seek qualitative, implicit, series, or numerical solutions.

Common trap

The phrase “solve the ODE” can mean several legitimate things: produce an explicit formula, produce an implicit relation, characterize a solution by an integral, give a convergent series, prove existence/uniqueness, describe qualitative behavior, or compute a numerical

approximation. Elementary closed form is only one possibility.

Part V

Exercises and Complete Solutions

Cumulative Exercises: Easy to Hard

22.1 Level A: Foundations

Exercises

- A1.** Find the domain and describe the level curves of $f(x, y) = \sqrt{16 - x^2 - 4y^2}$.
- A2.** Compute f_x, f_y for $f = x^2e^y + y \sin x$.
- A3.** Find the tangent plane to $z = x^2 + xy + y^2$ at $(1, 1)$.
- A4.** Find the gradient of $f = x^2 + 3y^2$ at $(1, -1)$ and the maximum rate of increase.
- A5.** Compute the directional derivative of $f = xy$ at $(2, 1)$ in the direction $(3, 4)$.
- A6.** Find dy/dx implicitly for $x^2 + xy + y^2 = 3$.
- A7.** Find the quadratic Taylor polynomial of $e^x \cos y$ at $(0, 0)$.
- A8.** Evaluate $\int_0^2 (2x + 1)dx$ from an antiderivative.
- A9.** Evaluate $\int 3x^2 \cos(x^3)dx$ by substitution.
- A10.** Evaluate $\int xe^x dx$ by parts.
- A11.** Compute $\int_0^1 \int_0^2 (x + y)dy dx$.
- A12.** Solve $y' = 2xy$, $y(0) = 1$.
- A13.** Solve $y' - y = x$.
- A14.** Solve $y'' - 5y' + 6y = 0$.
- A15.** Perform two Euler steps of size $h = 0.1$ for $y' = y + t$, $y(0) = 1$.

22.2 Level B: Intermediate

Exercises

- B1.** Determine whether $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^4+y^2}$ exists.
- B2.** Prove directly that $f(x, y) = x^2 + xy + 2y^2$ is differentiable at every point.
- B3.** For $f = x^2y + y^3$, find all critical points and classify them.

- B4.** Find extrema of xy subject to $x^2 + 4y^2 = 4$.
- B5.** Reverse the order of $\int_0^1 \int_{x^2}^x f(x, y) dy dx$.
- B6.** Evaluate $\iint_{x^2+y^2 \leq 4} e^{x^2+y^2} dA$.
- B7.** Find the Jacobian for $x = u^2 - v^2$, $y = 2uv$.
- B8.** Solve $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$ implicitly.
- B9.** Solve the Bernoulli equation $y' + y = xy^2$.
- B10.** Solve $y'' + 4y = \cos 2x$ and identify resonance.
- B11.** Solve $x^2 y'' - 3xy' + 4y = 0$ for $x > 0$.
- B12.** Use Euler's method with $h = 0.25$ to approximate $y(1)$ for $y' = y$, $y(0) = 1$, and compare to e .

22.3 Level C: Hard and proof-oriented

Exercises

- C1.** Show that if f is differentiable at $\mathbf{a}$ and has a local extremum there, then $Df(\mathbf{a}) = 0$.
- C2.** Let $f(x, y) = \frac{x^2 y}{x^4 + y^2}$ away from the origin and 0 at the origin. Investigate continuity and both coordinate partial derivatives at the origin.
- C3.** Prove that a positive definite Hessian at a critical point implies a strict local minimum using Taylor's theorem and eigenvalue bounds.
- C4.** Derive the polar Jacobian r geometrically and algebraically, and explain why omitting it gives dimensionally incorrect area near large r .
- C5.** Derive a third-order Taylor polynomial for $\log(1 + x + y)$ at $(0, 0)$ and specify a region where the underlying one-variable logarithm series converges absolutely.
- C6.** Solve the implicit ODE $y' = (x + y)/(x - y)$ by a homogeneous substitution and leave the answer in a clean implicit form.
- C7.** For $y' = y^{1/3}$, $y(0) = 0$, determine whether uniqueness holds and construct more than one solution.
- C8.** Prove Abel's identity for the Wronskian.
- C9.** Derive variation of parameters from the ansatz $y_p = u_1 y_1 + u_2 y_2$.
- C10.** For $y' = -10y$, determine the largest Euler step producing decaying numerical solutions. Explain what happens at and beyond the boundary.

Complete Solutions

23.1 Solutions to Level A

A1. The square root requires $16 - x^2 - 4y^2 \geq 0$, so the domain is the ellipse $x^2 + 4y^2 \leq 16$. A level $f = c \geq 0$ gives $x^2 + 4y^2 = 16 - c^2$, again ellipses.

A2.

$$f_x = 2xe^y + y \cos x, \quad f_y = x^2 e^y + \sin x.$$

A3. $f(1, 1) = 3$, $f_x = 2x + y$, $f_y = x + 2y$, so both partials are 3 at $(1, 1)$. Hence

$$z = 3 + 3(x - 1) + 3(y - 1) = 3x + 3y - 3.$$

A4. $\nabla f = (2x, 6y)$, so at $(1, -1)$ it is $(2, -6)$. Maximum rate is

$$\|(2, -6)\| = \sqrt{40} = 2\sqrt{10}.$$

A5. Unit direction is $(3/5, 4/5)$. Since $\nabla f = (y, x)$, at $(2, 1)$ it is $(1, 2)$. Therefore

$$D_{\mathbf{u}}f = (1, 2) \cdot (3/5, 4/5) = 11/5.$$

A6. Differentiate:

$$2x + y + xy' + 2yy' = 0.$$

Thus

$$y' = -\frac{2x + y}{x + 2y}.$$

A7. From the derivatives already developed,

$$T_2(x, y) = 1 + x + \frac{x^2 - y^2}{2}.$$

A8.

$$\int_0^2 (2x + 1)dx = [x^2 + x]_0^2 = 6.$$

A9. Let $u = x^3$, $du = 3x^2 dx$:

$$\sin(x^3) + C.$$

A10. Let $u = x$, $dv = e^x dx$:

$$e^x(x - 1) + C.$$

A11.

$$\int_0^1 \left[xy + \frac{y^2}{2} \right]_0^2 dx = \int_0^1 (2x + 2) dx = 3.$$

A12. Separate:

$$\frac{dy}{y} = 2x dx, \quad \ln |y| = x^2 + C, \quad y = Ce^{x^2}.$$

Using $y(0) = 1$ gives

$$\boxed{y = e^{x^2}}.$$

A13. Standard form $y' - y = x$ has integrating factor e^{-x} . Thus

$$(e^{-x}y)' = xe^{-x}.$$

Integration by parts gives $\int xe^{-x} dx = -(x+1)e^{-x}$, so

$$\boxed{y = Ce^x - x - 1}.$$

A14. Characteristic equation $r^2 - 5r + 6 = (r-2)(r-3) = 0$, so

$$\boxed{y = C_1 e^{2x} + C_2 e^{3x}}.$$

A15. $t_0 = 0, y_0 = 1$. First slope $f = 1$, so $y_1 = 1 + 0.1 = 1.1$ at $t_1 = 0.1$. Next slope is $1.1 + 0.1 = 1.2$, so $y_2 = 1.1 + 0.12 = \boxed{1.22}$ at $t = 0.2$.

23.2 Solutions to Level B

B1. Along $y = 0$ the function is 0. Along $y = x^2$,

$$\frac{x^2 x^2}{x^4 + x^4} = \frac{1}{2}.$$

Different path limits disagree, so the limit does not exist.

B2. This polynomial has continuous partial derivatives everywhere:

$$f_x = 2x + y, \quad f_y = x + 4y.$$

By the continuous-partials theorem, it is differentiable everywhere. Directly, at (a, b) the remainder after subtracting the linear terms is

$$h^2 + hk + 2k^2,$$

and dividing by $\sqrt{h^2 + k^2}$ tends to zero because the numerator is $O(h^2 + k^2)$.**B3.** $f_x = 2xy$, $f_y = x^2 + 3y^2$. The second equation is a sum of nonnegative terms, so it vanishes only at $(0, 0)$. Hessian at the origin is the zero matrix, so the test is inconclusive. But

$$f = x^2 y + y^3 = y(x^2 + y^2),$$

which changes sign with y . Therefore $(0, 0)$ is a saddle-type critical point, not an extremum.**B4.** Let $f = xy$, $g = x^2 + 4y^2$. Equations:

$$y = 2\lambda x, \quad x = 8\lambda y.$$

For nonzero coordinates, $1 = 16\lambda^2$, so $\lambda = \pm 1/4$. This implies $y = \pm x/2$. Constraint gives $x^2 + x^2 = 4$, so $x = \pm\sqrt{2}$ and $y = \pm 1/\sqrt{2}$ with matching or opposite signs. Thus $\max xy = 1$, $\min xy = -1$.

B5. The region is $0 \leq x \leq 1$, $x^2 \leq y \leq x$. For fixed $y \in [0, 1]$, the inequalities become $y \leq x \leq \sqrt{y}$. Therefore

$$\int_0^1 \int_y^{\sqrt{y}} f(x, y) dx dy.$$

B6. Polar coordinates: $0 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$:

$$\int_0^{2\pi} \int_0^2 e^{r^2} r dr d\theta = 2\pi \cdot \frac{1}{2}(e^4 - 1) = \boxed{\pi(e^4 - 1)}.$$

B7.

$$J = \det \begin{pmatrix} 2u & -2v \\ 2v & 2u \end{pmatrix} = 4u^2 + 4v^2 = \boxed{4(u^2 + v^2)}.$$

B8. Here $M = 2xy + y^2$, $N = x^2 + 2xy$. Since $M_y = 2x + 2y = N_x$, exact. Integrating M in x :

$$F = x^2y + xy^2 + g(y).$$

Matching $F_y = N$ gives $g' = 0$. Hence

$$\boxed{x^2y + xy^2 = C}.$$

B9. Equation $y' + y = xy^2$. Divide by y^2 for nonzero solutions:

$$y^{-2}y' + y^{-1} = x.$$

Let $v = y^{-1}$, so $v' = -y^{-2}y'$. Then

$$-v' + v = x, \quad v' - v = -x.$$

Integrating factor e^{-x} gives

$$(e^{-x}v)' = -xe^{-x}.$$

Integrating, $e^{-x}v = (x+1)e^{-x} + C$, so $v = x+1 + Ce^x$. Therefore

$$\boxed{y = \frac{1}{x+1+Ce^x}}.$$

Also $y = 0$ is an equilibrium solution lost by division.

B10. Homogeneous solution is $C_1 \cos 2x + C_2 \sin 2x$. Because the forcing $\cos 2x$ is resonant, try $y_p = Ax \sin 2x$. Substitution yields $4A \cos 2x = \cos 2x$, so $A = 1/4$. Thus

$$\boxed{y = C_1 \cos 2x + C_2 \sin 2x + \frac{x}{4} \sin 2x}.$$

B11. Try $y = x^m$. Then

$$m(m-1) - 3m + 4 = 0 \Rightarrow m^2 - 4m + 4 = (m-2)^2 = 0.$$

Repeated root $m = 2$ gives

$$\boxed{y = x^2(C_1 + C_2 \ln x)}, \quad x > 0.$$

B12. Euler with $h = 1/4$ for $y' = y$ multiplies by $1 + h = 1.25$ each step. Four steps give

$$y_4 = (1.25)^4 = 2.44140625.$$

Exact $e \approx 2.71828$, so error is about 0.27687.

23.3 Solutions to Level C

C1. At a differentiable local extremum, fix any vector $\mathbf{u}$ and define $g(t) = f(\mathbf{a} + t\mathbf{u})$. Then $t = 0$ is a local extremum of the one-variable differentiable function g , so $g'(0) = 0$. By the chain rule,

$$g'(0) = Df(\mathbf{a})(\mathbf{u}).$$

This holds for every $\mathbf{u}$, so the linear map $Df(\mathbf{a})$ is identically zero, equivalently $\nabla f(\mathbf{a}) = 0$.

C2. Along $y = x^2$, the function is $1/2$, while at the origin it is 0 , so it is not continuous. For $f_x(0, 0)$, set $y = 0$: $f(h, 0) = 0$, so the quotient is zero. For $f_y(0, 0)$, set $x = 0$: $f(0, k) = 0$, so that quotient is also zero. Thus both coordinate partial derivatives exist and vanish, yet the function is not continuous or differentiable.

C3. If H is positive definite and symmetric, its smallest eigenvalue $\lambda_{\min}$ is positive, and

$$\mathbf{h}^T H \mathbf{h} \geq \lambda_{\min} \|\mathbf{h}\|^2.$$

At a critical point, Taylor gives

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) = \frac{1}{2} \mathbf{h}^T H \mathbf{h} + o(\|\mathbf{h}\|^2).$$

For sufficiently small nonzero $\mathbf{h}$, the remainder has magnitude at most $\lambda_{\min} \|\mathbf{h}\|^2 / 4$. Therefore

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) \geq \frac{\lambda_{\min}}{4} \|\mathbf{h}\|^2 > 0.$$

So $\mathbf{a}$ is a strict local minimum.

C4. Algebraically the determinant is r . Geometrically, a small radial increment has length dr , while a small angular increment sweeps arc length approximately $r d\theta$. The tiny sector area is therefore $r dr d\theta$ to first order. Omitting r would claim equal area for angular cells near the origin and far away, even though the latter have proportionally longer arcs.

C5. Let $u = x + y$. Since

$$\log(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} + O(u^4),$$

we have

$$T_3 = x + y - \frac{(x + y)^2}{2} + \frac{(x + y)^3}{3}.$$

The power series converges absolutely for $|u| < 1$, so a natural region is $|x + y| < 1$.

C6. The ODE

$$y' = \frac{x + y}{x - y}$$

is homogeneous. Set $v = y/x$, so $y' = v + xv'$. Then

$$v + xv' = \frac{1 + v}{1 - v}.$$

Thus

$$xv' = \frac{1 + v - v + v^2}{1 - v} = \frac{1 + v^2}{1 - v},$$

so

$$\frac{1 - v}{1 + v^2} dv = \frac{dx}{x}.$$

Integrate:

$$\arctan v - \frac{1}{2} \ln(1 + v^2) = \ln|x| + C.$$

Substitute $v = y/x$. One clean implicit form is

$$\arctan(y/x) - \frac{1}{2} \ln(x^2 + y^2) = C,$$

valid on appropriate regions where a branch of $\arctan(y/x)$ is fixed.

C7. The function $f(y) = y^{1/3}$ is not locally Lipschitz at zero. One solution is $y = 0$. Separating for positive y gives

$$y^{-1/3} dy = dx \Rightarrow \frac{3}{2} y^{2/3} = x - C.$$

Thus after a waiting time $a \geq 0$ we may define

$$y_a(x) = \begin{cases} 0, & x \leq a, \\ \left(\frac{2}{3}(x-a)\right)^{3/2}, & x > a. \end{cases}$$

These are distinct solutions sharing $y(0) = 0$ when $a \geq 0$. Hence uniqueness fails.

C8. This is the Abel identity proof from the chapter: differentiate $W = y_1 y_2' - y_2 y_1'$, use $y_i'' = -p y_i' - q y_i$, cancel the q terms, obtain $W' = -pW$, and solve to get $W = C e^{-\int p dx}$.

C9. Start with $y_p = u_1 y_1 + u_2 y_2$. Differentiate:

$$y_p' = u_1' y_1 + u_2' y_2 + u_1 y_1' + u_2 y_2'.$$

Impose $u_1' y_1 + u_2' y_2 = 0$ so $y_p' = u_1 y_1' + u_2 y_2'$. Then

$$y_p'' = u_1' y_1' + u_2' y_2' + u_1 y_1'' + u_2 y_2''.$$

Substitute into the normalized equation. The terms multiplied by u_1, u_2 vanish because y_1, y_2 solve the homogeneous equation. We obtain

$$u_1' y_1' + u_2' y_2' = g.$$

Together with the imposed equation, Cramer's rule yields

$$u_1' = -\frac{y_2 g}{W}, \quad u_2' = \frac{y_1 g}{W}.$$

C10. Euler gives $y_{n+1} = (1 - 10h)y_n$. Decay requires

$$|1 - 10h| < 1,$$

so

$$0 < h < 0.2.$$

At $h = 0.2$, the multiplier is -1 , so the magnitude does not decay and signs alternate. For $h > 0.2$, the magnitude of the multiplier exceeds 1, so numerical values grow even though the exact solution e^{-10t} decays.

Part VI

Additional Proofs and Worked ODE Atlas

Proof Toolkit Behind the Main Theorems

24.1 The contraction mapping theorem

The existence proof for Picard iteration used a fundamental fixed-point theorem. Here is the complete argument in the form needed for ODEs.

Theorem 24.1 (Contraction mapping theorem). *Let (X, d) be a complete metric space and let $T : X \rightarrow X$ satisfy*

$$d(Tu, Tv) \leq q d(u, v)$$

for all $u, v \in X$, where $0 < q < 1$. Then T has a unique fixed point u_ , and the iteration $u_{n+1} = T(u_n)$ converges to it from every starting point.*

Proof

Choose any u_0 and define $u_{n+1} = T(u_n)$. Repeated use of the contraction property gives

$$d(u_{n+1}, u_n) \leq q^n d(u_1, u_0).$$

For $m > n$, the triangle inequality gives

$$d(u_m, u_n) \leq \sum_{k=n}^{m-1} d(u_{k+1}, u_k) \leq d(u_1, u_0) \sum_{k=n}^{m-1} q^k \leq \frac{q^n}{1-q} d(u_1, u_0).$$

The right side tends to zero as $n \rightarrow \infty$, uniformly in $m > n$. Therefore (u_n) is Cauchy. Completeness of X gives a limit $u_n \rightarrow u_* \in X$.

Because T is Lipschitz, it is continuous. Hence

$$T(u_*) = T(\lim u_n) = \lim T(u_n) = \lim u_{n+1} = u_*.$$

So u_* is a fixed point. If v_* is another fixed point,

$$d(u_*, v_*) = d(Tu_*, Tv_*) \leq q d(u_*, v_*).$$

Since $q < 1$, this forces $d(u_*, v_*) = 0$, so $u_* = v_*$. The same estimates show every iteration converges to this unique point.

24.2 Gronwall's inequality

Theorem 24.2 (Integral Gronwall inequality). *Suppose u is continuous and nonnegative on $[a, b]$ and*

$$u(t) \leq A + B \int_a^t u(s) ds,$$

where $A, B \geq 0$. Then

$$u(t) \leq Ae^{B(t-a)}.$$

Proof

Define

$$v(t) = A + B \int_a^t u(s) ds.$$

Then $u \leq v$ and

$$v'(t) = Bu(t) \leq Bv(t).$$

If $A > 0$, then $v > 0$ and

$$\frac{v'}{v} \leq B.$$

Integrating from a to t gives

$$\log v(t) - \log v(a) \leq B(t - a).$$

Since $v(a) = A$,

$$v(t) \leq Ae^{B(t-a)}.$$

Because $u \leq v$, the result follows. The case $A = 0$ follows by replacing A with an arbitrary $\varepsilon > 0$ and taking $\varepsilon \downarrow 0$.

This is the engine behind many stability and uniqueness estimates: a quantity that can only grow proportionally to its accumulated past cannot grow faster than an exponential.

24.3 One-variable Taylor theorem with integral remainder

Theorem 24.3. *If g has $n + 1$ continuous derivatives on an interval containing a and x , then*

$$g(x) = \sum_{k=0}^n \frac{g^{(k)}(a)}{k!} (x - a)^k + \frac{1}{n!} \int_a^x (x - t)^n g^{(n+1)}(t) dt.$$

Proof

Start with the Fundamental Theorem:

$$g(x) = g(a) + \int_a^x g'(t_1) dt_1.$$

Apply the same identity to $g'(t_1)$:

$$g'(t_1) = g'(a) + \int_a^{t_1} g''(t_2) dt_2.$$

Substituting gives

$$g(x) = g(a) + g'(a)(x - a) + \int_a^x \int_a^{t_1} g''(t_2) dt_2 dt_1.$$

Changing the order in the triangular integration region turns the double integral into

$$\int_a^x (x-t)g''(t)dt.$$

Repeating this process n times gives

$$g(x) = \sum_{k=0}^n \frac{g^{(k)}(a)}{k!} (x-a)^k + \frac{1}{n!} \int_a^x (x-t)^n g^{(n+1)}(t)dt.$$

This proof shows Taylor's theorem as repeated accumulation of derivative information rather than a mysterious polynomial trick.

24.4 Why the determinant measures area scaling

For a linear map

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the unit square is sent to the parallelogram spanned by the columns (a, c) and (b, d) . Its signed area is

$$ad - bc = \det A.$$

Proof

Assume first $a \neq 0$. The parallelogram has base vector (a, c) . Apply the shear

$$S = \begin{pmatrix} 1 & 0 \\ -c/a & 1 \end{pmatrix}.$$

This shear preserves horizontal base lengths and vertical heights, hence area. It sends the first column to $(a, 0)$ and the second to $(b, d - bc/a)$. The resulting parallelogram has signed area

$$a \left(d - \frac{bc}{a} \right) = ad - bc.$$

If $a = 0$, interchange axes or use continuity. Taking absolute value gives ordinary geometric area. Thus the determinant is exactly the local area-stretching factor used in change of variables.

Worked ODE Atlas: Recognize the Structure

25.1 Example 1: separable growth and decay

Solve

$$y' = (1 + x^2)(1 - y^2).$$

Separate:

$$\frac{dy}{1 - y^2} = (1 + x^2)dx.$$

Using partial fractions,

$$\int \frac{dy}{1 - y^2} = \frac{1}{2} \log \left| \frac{1 + y}{1 - y} \right|.$$

Hence

$$\frac{1}{2} \log \left| \frac{1 + y}{1 - y} \right| = x + \frac{x^3}{3} + C.$$

This implicit relation is already a correct general solution on intervals avoiding $y = \pm 1$. The equilibrium solutions $y \equiv 1$ and $y \equiv -1$ must also be added because separation divided by $1 - y^2$.

25.2 Example 2: logistic equation

The population model

$$P' = rP \left(1 - \frac{P}{K} \right)$$

is separable. Partial fractions give

$$\frac{K}{P(K - P)} = \frac{1}{P} + \frac{1}{K - P}.$$

Thus

$$\log \frac{P}{K - P} = rt + C,$$

and

$$\boxed{P(t) = \frac{K}{1 + Ae^{-rt}}.}$$

The equilibrium solutions are $P = 0$ and $P = K$. For $0 < P < K$, the population increases; for $P > K$, it decreases toward K .

25.3 Example 3: linear mixing model

A tank contains V liters and receives solute at constant concentration c_{in} with equal inflow/outflow rate q . If $S(t)$ is solute amount,

$$S' = qc_{in} - q\frac{S}{V}.$$

This is linear:

$$S' + \frac{q}{V}S = qc_{in}.$$

The solution is

$$S(t) = Vc_{in} + (S_0 - Vc_{in})e^{-qt/V}.$$

The physical meaning is transparent: the amount exponentially approaches the equilibrium Vc_{in} .

25.4 Example 4: exact equation with implicit answer

Solve

$$(ye^x + 2x)dx + (e^x + 2y)dy = 0.$$

Here

$$M_y = e^x = N_x,$$

so the equation is exact. Integrate M in x :

$$F = ye^x + x^2 + g(y).$$

Then $F_y = e^x + g'(y) = e^x + 2y$, so $g = y^2$. Therefore

$$ye^x + x^2 + y^2 = C.$$

Trying to isolate y would add algebra without adding mathematical value. The implicit form is natural.

25.5 Example 5: Bernoulli equation

Solve

$$y' + \frac{1}{x}y = x^2y^3, \quad x > 0.$$

Set $v = y^{-2}$. Since $v' = -2y^{-3}y'$, multiply the ODE by y^{-3} :

$$y^{-3}y' + \frac{1}{x}y^{-2} = x^2.$$

Thus

$$-\frac{1}{2}v' + \frac{1}{x}v = x^2,$$

or

$$v' - \frac{2}{x}v = -2x^2.$$

The integrating factor is x^{-2} . Hence

$$(x^{-2}v)' = -2,$$

so

$$v = x^2(C - 2x).$$

Therefore

$$y = \pm \frac{1}{x\sqrt{C - 2x}}$$

on intervals where the expression is real, together with the lost solution $y = 0$.

25.6 Example 6: homogeneous substitution

Solve

$$y' = 1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2.$$

Set $v = y/x$, so $y' = v + xv'$. Then

$$v + xv' = 1 + v + v^2,$$

so

$$xv' = 1 + v^2.$$

Therefore

$$\frac{dv}{1 + v^2} = \frac{dx}{x},$$

and

$$\boxed{\arctan(y/x) = \log|x| + C.}$$

This is a clean implicit answer.

25.7 Example 7: second-order constant coefficients

Solve

$$y'' + 2y' + 5y = 0.$$

The characteristic roots are

$$r = -1 \pm 2i.$$

Hence

$$\boxed{y = e^{-x}(C_1 \cos 2x + C_2 \sin 2x).}$$

The geometry is a decaying oscillation: real part -1 controls exponential damping and imaginary part 2 controls angular frequency.

25.8 Example 8: polynomial forcing

Solve

$$y'' - y = x^2.$$

Homogeneous solution:

$$y_h = C_1 e^x + C_2 e^{-x}.$$

Try $y_p = Ax^2 + Bx + C$. Then

$$2A - (Ax^2 + Bx + C) = x^2.$$

Matching coefficients gives $A = -1$, $B = 0$, and $C = -2$. Thus

$$\boxed{y = C_1 e^x + C_2 e^{-x} - x^2 - 2.}$$

25.9 Example 9: resonance

Solve

$$y'' + 9y = \sin 3x.$$

The forcing frequency equals the natural frequency 3. Because $\sin 3x$ is already in the homogeneous solution, multiply the usual trial by x . A convenient trial is

$$y_p = Ax \cos 3x.$$

Substitution gives $-6A \sin 3x = \sin 3x$, hence $A = -1/6$. Therefore

$$y = C_1 \cos 3x + C_2 \sin 3x - \frac{x}{6} \cos 3x.$$

25.10 Example 10: reduction when y is absent

Solve

$$y'' = (y')^2.$$

Let $v = y'$. Then

$$v' = v^2.$$

Separate:

$$\frac{dv}{v^2} = dx,$$

so

$$-\frac{1}{v} = x + C_1, \quad v = -\frac{1}{x + C_1}.$$

Integrate again:

$$y = -\log |x + C_1| + C_2.$$

Also $v = 0$ gives the constant solutions $y = C_2$, which were lost by dividing by v^2 .

25.11 Example 11: reduction when x is absent

Solve

$$y'' = yy'.$$

Let $v(y) = y'$. Then

$$y'' = v \frac{dv}{dy}.$$

The equation becomes

$$v \frac{dv}{dy} = yv.$$

One branch is $v = 0$, giving constant solutions. Otherwise divide by v :

$$\frac{dv}{dy} = y,$$

so

$$v = \frac{1}{2}y^2 + C_1.$$

Thus the original second-order equation reduces to the first-order separable equation

$$y' = \frac{1}{2}y^2 + C_1.$$

The final integral depends on the sign of C_1 and may involve logarithms or tangent functions. The important lesson is the order reduction.

25.12 Example 12: when symbolic methods stop

Consider

$$y' = y^2 - x, \quad y(0) = 0.$$

It is not separable, not linear, not exact in an immediately useful way, and not Bernoulli. There is no reason to expect an elementary formula. A mathematically complete response can instead establish local existence and uniqueness, analyze the slope field, construct a power series, or approximate numerically. For example, Euler gives

$$y_{n+1} = y_n + h(y_n^2 - t_n).$$

Recognizing when to stop forcing elementary algebra is part of solving differential equations well.

Part VII

Reference Material

Formula and Concept Sheet

26.1 Multivariable derivative

$$Df(a, b)(h, k) = f_x(a, b)h + f_y(a, b)k = \nabla f(a, b) \cdot (h, k).$$

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}.$$

$$D(f \circ g) = Df(g)Dg.$$

26.2 Implicit differentiation

If $F(x, y) = 0$ and $F_y \neq 0$,

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

26.3 Second-order Taylor

$$f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + \nabla f^T \mathbf{h} + \frac{1}{2} \mathbf{h}^T H_f \mathbf{h} + o(\|\mathbf{h}\|^2).$$

26.4 Second derivative test

$$D = f_{xx}f_{yy} - f_{xy}^2.$$

$D > 0$ gives min/max according to sign of f_{xx} ; $D < 0$ gives saddle.

26.5 Integral identities

$$\int_a^b f = F(b) - F(a), \quad F' = f.$$

$$\int f(g(x))g'(x)dx = \int f(u)du.$$

$$\int u dv = uv - \int v du.$$

$$\iint_R f \, dA = \int \int f \, dy \, dx.$$

$$dA = \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv.$$

Polar:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad dA = r \, dr \, d\theta.$$

26.6 First-order ODEs

Separable:

$$y' = g(x)h(y) \Rightarrow \int \frac{dy}{h(y)} = \int g(x)dx + C.$$

Linear:

$$y' + py = q, \quad \mu = e^{\int p dx}, \quad y = \mu^{-1} \left(\int \mu q \, dx + C \right).$$

Exact:

$$Mdx + Ndy = 0, \quad M_y = N_x, \quad F_x = M, F_y = N, \quad F = C.$$

26.7 Second-order constant coefficients

$$ay'' + by' + cy = 0 \Rightarrow ar^2 + br + c = 0.$$

Distinct roots: $C_1 e^{r_1 x} + C_2 e^{r_2 x}$. Repeated root: $(C_1 + C_2 x) e^{rx}$. Complex roots $\alpha \pm i\beta$: $e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$.

26.8 Euler method

$$t_{n+1} = t_n + h, \quad y_{n+1} = y_n + hf(t_n, y_n).$$

Local truncation error: $O(h^2)$. Global error: $O(h)$ under standard hypotheses.

Final Mental Map

The entire course can be compressed into a chain of ideas:

$\text{function} \rightarrow \text{local change} \rightarrow \text{linearization} \rightarrow \text{quadratic correction} \rightarrow \text{accumulation} \rightarrow \text{evolution}.$

For one variable, the derivative is a number. For many variables, the derivative is a linear map. The gradient represents that map for scalar outputs. The Hessian is the next-order correction. Taylor's theorem turns derivatives into local polynomial models. Integrals reconstruct totals from infinitesimal contributions, and the Fundamental Theorem connects accumulation back to differentiation. Substitution and integration by parts are the chain rule and product rule read backward. Multiple integrals repeat the accumulation idea over area or volume, with Jacobians accounting for local coordinate stretching. Differential equations then prescribe derivative information and ask us to reconstruct the unknown function, exactly, implicitly, qualitatively, or numerically.

The deepest habit to keep is this: whenever you see a formula, ask which local approximation or one-variable differentiation rule is hiding behind it. Most of the subject becomes reconstructible once those few structural ideas are understood.

Suggested study sequence

A strong route through the book is:

1. Chapters 1–4: functions, limits, partial derivatives, total differentiability.
2. Chapters 5–9: gradient, chain rule, implicit functions, Taylor theory, optimization.
3. Chapters 10–14: integral meaning, FTC, substitution, integration by parts, double integrals, Jacobians.
4. Chapters 15–22: ODE meaning, first-order methods, existence/uniqueness, second-order methods, oscillations, Euler method.
5. Work Level A exercises without notes, then Level B, then Level C.
6. Return to the reference sheet only after you can explain where each formula comes from.